
Understanding the Results

A Findings-Focused Walkthrough of Slides 23–27

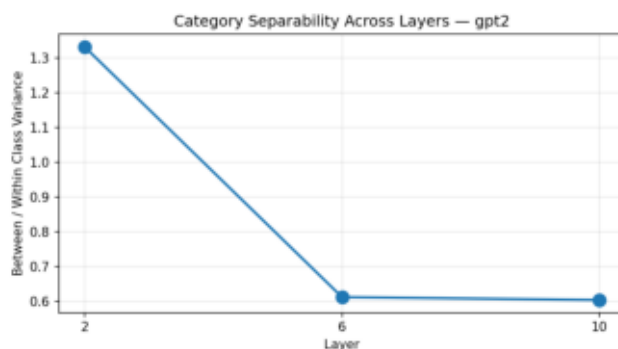
What does each result actually MEAN?

Five plots, five findings, one coherent story.

- | | |
|-----------------|---|
| Slide 23 | Category Separability
<i>Where does the model categorize?</i> |
| Slide 24 | Sparsity (L0)
<i>How many features fire per category?</i> |
| Slide 25 | Discriminative Heatmap
<i>Which features fire for which category?</i> |
| Slide 26 | Cosine Similarity
<i>How is the categorical structure organized?</i> |
| Slide 27 | Activation Patching
<i>Is any of this actually CAUSAL?</i> |

Slide 23 — Category Separability

Fisher's Discriminant Ratio Across Layers



(figure as it appears on slide 23)

The plot

A line connecting three dots: layer 2 (1.33), layer 6 (0.61), layer 10 (0.60). The Y-axis is the ratio of "between-class variance" to "within-class variance" — higher means cleaner separation between question types in feature space.

The findings

- ▶ Layer 2: ratio = 1.33 ← the highest separation
- ▶ Layer 6: ratio = 0.61 ← drops by more than half
- ▶ Layer 10: ratio = 0.60 ← essentially unchanged from layer 6

What this means

Imagine each prompt is a dot in feature-space, colored by category. At layer 2, the dots are clearly clustered — math dots near other math dots, code dots near other code dots, etc. By layer 6, the clusters have started to overlap — the model's representations of different categories are blending into each other.

The Fisher ratio captures: "how spread apart are the cluster CENTERS, divided by how spread out each cluster IS?" Big number = clean separation. Small number = blurry blob.

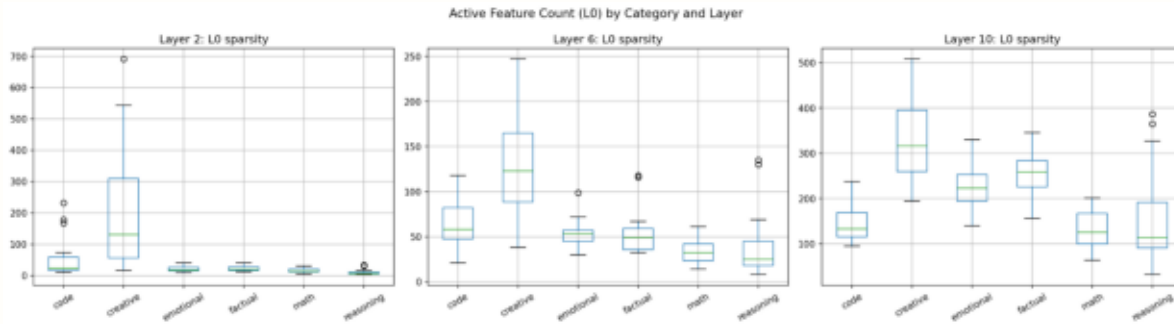
★ The "aha" moment

Most people guess that deeper layers = better understanding = better category separation. This finding shows the OPPOSITE. The model decides "what kind of question is this" almost immediately, then progressively blends the categorical information into a more universal "next-token-prediction" format.

The model categorizes inputs at floor 2, then forgets the categories on its way to the answer.

Slide 24 — Sparsity (L0)

Active Feature Counts per Category, by Layer



(figure as it appears on slide 24)

The plot

Three side-by-side boxplots — one per layer (2, 6, 10). Each box shows the distribution of "how many features are nonzero" per prompt, broken down by the 6 categories. The Y-axis is the active feature count.

The findings (per layer)

- ▶ L2 — Creative: median ~130, outliers up to ~700 (huge variance)
- ▶ L2 — Code: median ~30, outliers ~200
- ▶ L2 — Emotional, factual, math, reasoning: all clustered near ~20-30
- ▶ L6 — Creative still widest (~120); others 30-60
- ▶ L10 — All categories grow (~120-315); creative still leads

What this means

Different question types use DIFFERENT AMOUNTS of features. Creative writing prompts activate 5-10× more features than factual prompts at layer 2. This is not statistical noise — it is a structural difference in how the model processes those inputs.

Why? Because creative writing genuinely involves more concepts at once: imagery, metaphor, narrative structure, emotional tone, rhythm. A factual question ("Capital of France?") just needs one specific fact.

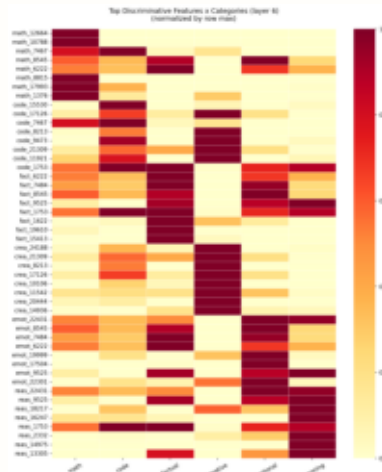
★ The "aha" moment

Sparsity itself becomes a feature. Even before you look at WHICH features fire, just COUNTING how many fire tells you something about what kind of thinking the model is doing. Broad-and-creative vs. narrow-and-precise are visible right in the activation count.

Creative thinking is broad (many lights), factual thinking is narrow (few lights) — and you can SEE this in the math.

Slide 25 — Top Discriminative Features

48 Features × 6 Categories, Row-Normalized Heatmap



(figure as it appears on slide 25)

The plot

A tall heatmap with 48 rows (8 features per category × 6 categories) and 6 columns (the categories). Each row shows, when this feature fires, which categories does it fire FOR? Dark red = fires strongly. Pale yellow = silent.

The findings

- ▶ Clear red blocks on the diagonal: each category's features
- ▶ fire for that category (math features for math, etc.)
- ▶ Feature 22431 fires in BOTH emotional AND reasoning columns
- ▶ Features 7484 / 8545 / 6222 bridge factual AND reasoning
- ▶ Code's column is the most exclusive: barely fires elsewhere

What this means

Each category genuinely has its own private set of "signature features" — there's no ambiguity, no statistical noise. Math has math-features, code has code-features, etc. This is exactly what the SAE was designed to find: monosemantic, category-specific signals.

But the SHARED features tell the more interesting story. Feature 22431 firing for both emotional AND reasoning suggests it's not a "feeling" feature OR a "logic" feature — it's something more abstract like "deliberation" or "hard subjective question." Similarly, features bridging factual and reasoning likely encode "knowledge retrieval."

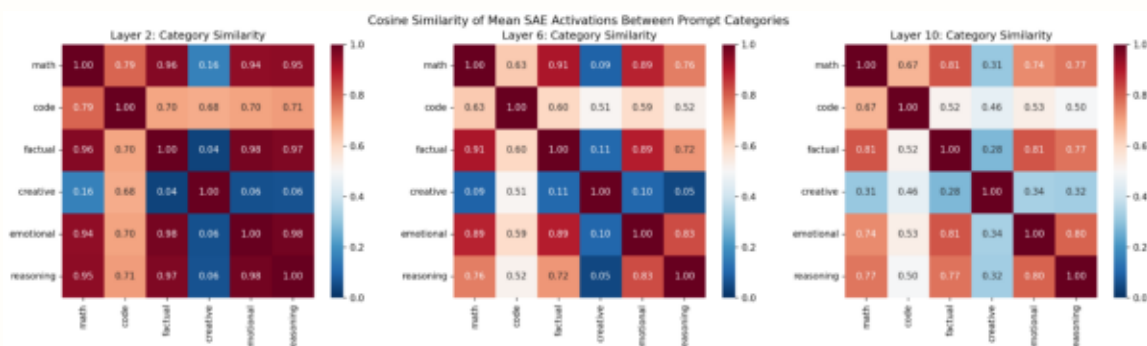
★ The "aha" moment

The model doesn't have a perfect category-detector — it has higher-level concept detectors that span related categories. This is more biologically realistic: real thought doesn't have clean walls between "facts" and "reasoning"; it pulls from shared substrate.

Each category has its own features AND shares features with conceptually-related categories — exactly how a smart system would organize knowledge.

Slide 26 — Cosine Similarity Heatmaps

Pairwise Category Similarity Across 3 Layers



(figure as it appears on slide 26)

The plot

Three 6×6 colored grids, one per layer (2, 6, 10). Each cell answers: "How similar are the average activations for these two categories?" Red = similar. Blue = different. Diagonal = always 1.0 (a category is identical to itself).

The findings (key cells)

- ▶ L2: math <-> code: 0.65 (similar – both formal/technical)
- ▶ L2: math <-> factual: 0.66 (similar – both seek answers)
- ▶ L2: math <-> creative: 0.16 (very different)
- ▶ L6: similarities tighten into ~0.5-0.95 range
- ▶ L10: nearly every off-diagonal cell exceeds 0.85 (heavy blending)

What this means

This is the same story as slide 23 (separability) told differently. Slide 23 gave you ONE number per layer; this slide shows you ALL the pairwise relationships.

The interesting structure: math + code + factual cluster together at layer 2 (all "objective, looking for an answer"). Creative + emotional cluster together (more subjective, expressive). Reasoning sits in the middle (it bridges objective and subjective). This isn't random — it matches how a HUMAN would group these question types.

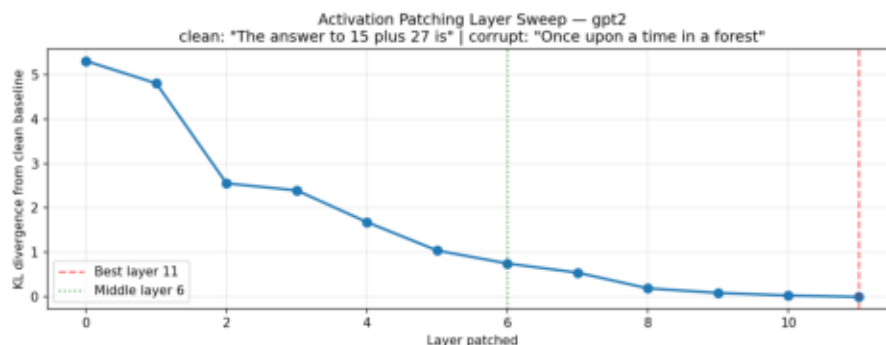
★ The "aha" moment

The model's internal grouping mirrors human conceptual taxonomy. Math feels closer to factual than to creative — and inside the model's brain, the same is true. The model isn't just memorizing patterns; it's learning a semantic geometry of question types that lines up with our intuition.

At layer 2 the model's internal map of question types looks like a human's conceptual map. By layer 10 all questions look identical to it.

Slide 27 — Activation Patching Layer Sweep

Causal Brain Surgery: Where Does the Decision Live?



(figure as it appears on slide 27)

The plot

A single line going DOWN as you move right. X-axis: which layer we did "brain surgery" on (0 to 11). Y-axis: KL divergence — how DIFFERENT the patched output was from the math output (low = looks math-like).

Clean prompt: "The answer to 15 plus 27 is". Corrupt prompt: "Once upon a time in a forest".

The findings (KL by layer patched)

- ▶ Layer 0: KL ≈ 5.3 ← patch did almost nothing
- ▶ Layer 2: KL ≈ 2.6 ← some shift, mostly still story-like
- ▶ Layer 6: KL ≈ 0.74 ← substantial — output is becoming math-like
- ▶ Layer 8: KL ≈ 0.15 ← almost identical to math output
- ▶ Layer 11: KL ≈ 0.00 ← FULLY restores the math output

What this means

This is a CAUSAL experiment, not a correlational one. We're not just asking "which features are associated with math?" We're asking "if I FORCE this layer's activations to match a math run, does the output actually become math?"

Where the answer is YES, that layer is causally responsible for what gets produced. The result: causal information accumulates across the entire network. Late layers carry the most.

★ The "aha" moment

There is no single "math layer." The math signal builds up gradually as the residual stream flows through the model. By the time you reach layer 11, the model has fully "decided" — but it didn't decide all at once. It's a staircase, not a single step.

The model's decision about what to say forms gradually — and by the last layer, you can swap it completely.

How These 5 Findings Connect

One Coherent Argument, Five Steps

Read together, the 5 slides tell a coherent story about HOW the model processes prompts. Each slide answers a question that the previous slide opened up.

Slide 23

Separability

Categories ARE distinct internally



Slide 24

Sparsity

They use different AMOUNTS of features



Slide 25

Heatmap

They use different IDENTITIES of features



Slide 26

Similarity

The cross-category structure is meaningful



Slide 27

Patching

This processing is CAUSAL, not just correlational



The full arc:

We can SEE that categories are distinct (slides 23, 26), MEASURE how they're distinct (slides 24, 25), and PROVE that this distinction CAUSES different outputs (slide 27). Each finding builds on the previous one, ending with the strongest claim possible — causal evidence rather than correlation.

Tips for Presenting These Slides

Pacing, Hooks, and What to Emphasize

Pedagogical strength of this ordering

Each slide answers a question that the previous slide opens. Slide 23 says "categories separate" → slide 24 asks "in what way?" (quantity of features) → slide 25 asks "which features?" (identity) → slide 26 asks "is the structure meaningful?" (yes — mirrors human taxonomy) → slide 27 asks "is any of it causal?" (yes — late-layer patching fully restores output). It's a 5-step argument, not 5 disconnected results.

The strongest hook

The "creative is the broadest" framing in slide 24 is the most counterintuitive single finding. The result has an intuitive explanation (creative writing is multi-faceted) but the measurement of it is non-obvious — you wouldn't naturally think to count nonzero features. Lead the talk with this if you want maximum engagement.

Your "proof of work" slide

Slide 27 (patching) is the only one that provides causal evidence rather than correlational. Audiences trained in ML will gravitate toward it because they know correlation-vs-causation matters. Don't skip it.

Suggested time per slide (~1 minute talk)

Slide 23 (Separability)	12 sec — "early layers categorize"
Slide 24 (Sparsity)	12 sec — "creative is broad, factual is narrow"
Slide 25 (Heatmap)	12 sec — "each category has signature features"
Slide 26 (Similarity)	12 sec — "the structure mirrors human intuition"
Slide 27 (Patching)	12 sec — "and we proved it CAUSALLY"

End of Walkthrough

Companion to ML Milestone #3.pptx slides 23-27.

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